

CM1205: Architecture and Operating Systems

STRUCTURED PORTFOLIO **(100%)**

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WEEK 1: INTRODUCTION

Reflection on Learning Materials

This was my first week with CM1205, and hence was more of an introduction to the module as in the first week we did not have a seminar discussion. The learning materials presented us with a history of how computers have developed over the years. What particularly caught my attention was the constant need for man to make their life much easier by using these small computers. It started with simple calculators, to devices that filled entire rooms which eventually have been made into small devices which can fit in the palm of our hand.

My favorite example of early computers presented to us was Napier's Bones (1617), which were developed to make life easier for people who needed to do multiplication calculations – hence, these computers would now be referred to as calculators. How this works was shown to us via a YouTube video, where it was shown how they used sticks to show what the result of multiplication was. This led to the development of other computers such as Pascaline (1642) and the slide rule (1621). However, these calculations were still nowhere near what computers are capable of doing nowadays. These calculators were at one point referred to as computers, whereas now computers have become far more powerful than calculations.

Following this, I learned about the development of the first computers from important figures such as Charles Babbage, "the father of computers" and Alan Turing, one of the most important people in computer history. Then, we took a look at the development of computer techniques such as Boolean mathematics and Von Neumann Architecture, which we have learned in a previous module. We finally ended on modern computers which are all based on the previously mentioned ideas presented to us just with more powerful hardware.

WEEK 2: STRUCTURE OF COMPUTERS

Reflection on Learning Materials

We began this week's learning materials by looking at Moore's law, which we have touched upon in a previous module, which states over time computers decrease in cost and increase in power at a rate of factor 2 per year. We then dove into other laws such as Nathan's first law of software and the Virtuous economic circle, which both show we adapt our computers to fit our needs. These factors show how we are constantly changing our computers to make our lives much easier. We are shown how these computers do so through the computer-level hierarchy, which goes from executable programs to circuits and binary logic. We were shown these through various diagrams to visualize what occurs during the computers' processes, focusing largely on the most general-purpose architecture Von Neumann which we have looked at in the previous week and during CM1101.

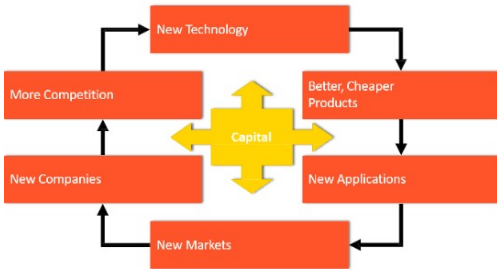


Figure 1 – this figure shows the Virtuous economic circle

Following this, we dove into more of the architecture within the computer, such as truth tables, logic gates, and Boolean algebra, which I

am already familiar with. However, I had to do my research on Transistors as Logic Gates as I was unaware of this concept, where I watched “Transistor Logic Gates - NAND, AND, OR, NOR” (The Organic Chemistry Teacher, 2022) to help me understand how transistors can represent these logic gates.

We were given the exercise based on these logic gates to test our knowledge and calculations, which I did not particularly struggle with, before being introduced to half and full adders which finishes this part on these logic gates. Then finally, we dove into binary, processors, and other hardware in computers, which are principles I am previously familiar with and have also had a decent introduction to within CM1101.

Reflection on Seminar Discussion

In this seminar, we discussed how we would deliver IT training for an elderly woman whom also requested to learn the history of computers. We discussed techniques in order to explain the best way for someone like her to understand, with the general idea of using analogies to break it down into something she is already familiar with, which based off of her age would be things generally around 40 years ago. We also would use her hobbies in order to explain this. For example, she enjoys gardening, so terms like planting files may be helpful to understand that the files can grow once we nurture them (make changes to the file). We also need to do this due to her lack of education as it is not further than secondary school which may in general imply, she is not as familiar with newer technology etc. As for explaining the history, we would explain how computers have gotten faster and to avoid confusion remove irrelevant details.

WEEK 3: OPERATING SYSTEMS

Reflection on Learning Materials

This week for CM1205, we used slides given to us in this week to learn about assembly language, a low-level language, and its uses. They presented us with facts about the language, how it is useful, and how it is less useful nowadays, pointing out that the source for these statements was more than likely heavily favoring assembly language. Within the slides, we were given audio clips with the lecturer discussing the topic at hand. Outside of the slides, we learned about different types of assembly languages, as well as common uses both nowadays and in history. It is mainly developed around the idea of coordinating the hardware on how to operate, which we now no longer need to do as much due to the fact we have high-level languages and compilers to do so for us.

I found it intriguing how assembly language can still be used today, despite the fact it is far easier not to. I agree that this is an extremely complicated way of programming, and in fact, it can be done for us to save time and reduce the large number of errors that will occur if we were to write it ourselves. However, I see the possibilities of reverse engineering this language to help develop complex software code such as malware and anti-virus. Furthermore, they are generally faster and take up less space with more capabilities. Despite this, these do not outweigh the negatives of assembly language.

Alongside facts, we had images to present to us the discussion about assembly language. We were also suggested to watch the German film Persischstunden (2020), which is related to how assembly language is a higher level of machine code, which I had not watched before being set this task. After watching it, I understood how assembly languages are related to machine code and its creation by Wolfgang Kohlhaase during the Second World War. After we had been presented with this, we were split into groups to discuss how to go about the task given to us. This will be further explained in the following passage.

Reflection on Seminar Discussion

Here, we discussed in the seminar how we would approach a disgruntled customer who has a complaint about the CPU the employee they are talking to told them they should get as the one they thought was better was named i7 as opposed to the i5 model which the employee suggested. We would need to explain to the customer, in terms that do not require the customer to have a huge understanding of computers that the i5 model is far newer, works significantly faster due to the number of cores, and that the price difference may be worth paying for the optimal machine. We also showed the suggestion of using websites such as UserBenchmark to show the customer what the processing power looks like to explain to them why the i5 may sound older but runs far better.

WEEK 4: ASSEMBLY LANGUAGE

Reflection on Learning Materials

In this week's C1205 learning, we learned about assembly code and machine code, starting with looking into ISA (Instruction Set Architecture) to understand Assembly Language. We were given slides and audio notes to explain the architecture of a computer. We went into detail about registers and other levels of the operating system, presented in Figure 2. What I found particularly interesting was how the Intel 8086 register was broken down. I found figure 3 a particularly interesting breakdown of the register, alongside the key as to explain what each two letters represent, as well as the breakdown of each location. For example, we can see that there are AX, BX, CX, and DX. Each of these holds 2 bits of data, meaning that 8 bits of data can be held within the general purpose. register, with the other 8 bits being reserved for indexing the locations of data.

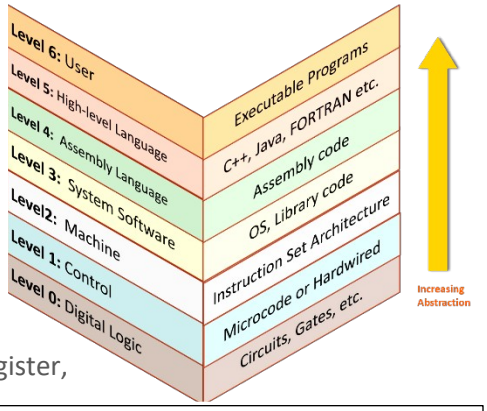


Figure 2, operating system levels.

Following the explanation of the 8086 16-bit register, we were shown the 32-bit register and suggested doing outside research on registers such as these to explain each detail was too complex for the slides. I found that it was far harder to comprehend the 32-bit Pentium Register than the previously mentioned 16-bit register. I

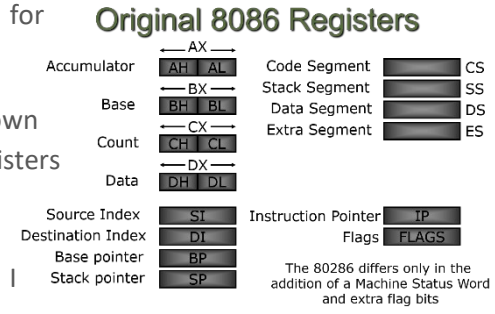


Figure 3, 8086 register.

used sources such as Princeton University to break down this register in greater detail, along with other sites to explain this and explain in greater detail flags.

Once we had done the slides, we were given a lab task, in which we were required to set up Visual Studio and examine how registers change when running basic assembly language code. We located where the registers were presented to the user and examined how they changed during code along with recording results between different exercises alongside how the variables and flags change throughout. I have little experience with assembly language; however, I have done research into Von Neuman Architecture and Little Man Computer before, hence I had a general understanding of how transferring data between registers worked. It was very interesting to see how assembly runs within the computer and how registers and flags work.

WEEK 5: ADDRESSING MODES

Reflection on Learning Materials

During this week's learning, we were required to take a look further at assembly language, specifically at the addressing modes used to begin. We looked at how they run and what they are used for, based on our knowledge from the previous weeks' work. I am somewhat familiar with these addressing modes before, however, there is nowhere near as much detail as we were taught here. We were shown this via slides and explanations.

Then, we were shown loops and jumps using assembly language. With this, we were shown pseudocode of sorts and examples of how they would run for assembly. Showing us how the code would be written in assembly, for me, was by far the greatest recourse we were given during this week's learning. Examples of this code can be seen below:

```
if (a < b) then X = 1 else X = 2;
  MOV EAX, A          ; Load value of A
  CMP EAX, B          ; Compare it with B
  JGE ElseCase        ; If A >= B then "else"
  MOV X, 1            ; Otherwise do "then"
  JMP EndIf           ; Jump to end of if
ElseCase:
  MOV X, 2            ; Do "else" case
EndIf:
```

```
  MOV ECX, 10       ; initialise counter
ForLoop:
  ADD X, ECX        ; do loop body
  LOOP ForLoop      ; repeat if ECX > 1
```

Following this, we were required to perform lab exercises, as we did in the previous week. Here, we were first required to run assembly on our machine once again and this time test the jump instructions. We had to use code given to us from the exercise sheet and execute it, before slightly altering the code to change the output and show how it changes the registers and flags. Following this, we had to repeat this for a loop before finally being tasked with combining two pieces of given code so that it reads from the keyboard and prints it on the screen. I find these hands-on tasks the most interesting and the best for me to learn as I particularly like solving logical coding problems.

WEEK 6: PROCEDURES AND SUBROUTINES

Reflection on Learning Materials

Throughout this week's learning, we looked into assembly code structures using slides as a resource. We explored the passing of parameters to procedures and functions, observing how they are executed. Understanding how stacks utilize parameter storage, particularly for addresses rather than data itself, was interesting to discover. Visual aids accompanied by audio explanations helped the explanation, especially when dissecting assembly code examples and observing register modifications. For instance, we examined passing parameters using global variables, which demonstrated how registers behave during data transmission. Additionally, we explored string manipulation in assembly, crucial for user interactions. We had a look at how this was done via the direction flags. Examples given to us along the way helped me understand how strings are dissected in assembly, highlighting the movements between registers. We were then shown the nine-string instructions and how they work. With these, I was able to understand the general idea of handling strings. The 9 instructions are as follows:

1) REP - Repeat
2) REPE (REPZ) - Repeat while equal (zero)
3) REPNE (REPNZ) - Repeat while not equal (NZ)
4) MOVS - Move byte or word string
5) MOVSB (MOVSW) - Move byte string (word string)
6) CMPS - Compare byte or word string
7) SCAS - Scan byte or word string
8) LODS - Load byte or word string
9) STOS - Store byte or word string

Once we had completed these tasks, we were tasked with a 3rd set of exercises, which involved combining two files that read and wrote whatever the user inputs after a series of set text. Then, we were required to create a subroutine. Creating subroutines proved relatively straightforward as it is a skill that was required with other pieces of work, such as object-orientated Java. However, creating and altering these pieces of code were my favorite activities to complete in the three previous weeks as they gave me an understanding of how assembly works.

WEEK 8: MEMORY, SECONDARY MEMORY, I/O, BUS

Reflection on Learning Materials

For this week, we dove into memory. This is a subject with which I am fairly familiar due to my previous experience at A level and whilst building my computer. Through the use of a presentation, we had a look at the parameters for the memory hierarchy, as presented on the right in Figure 4. Then, we took a quick look at the cache (again something I am fairly familiar with) and then secondary storage, specifically hard disk drives.

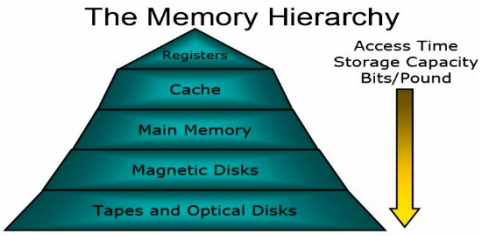


Figure 4

Here, I was able to learn how the design of these disks has changed over time, specifically how much data the sectors can hold with the old model it used to be that all segments held the same amount of data, whereas now they can hold more relative to size, of which increases the further out on the track you go. Following this, we took an in-depth look into data organization. We looked at FAT systems and clusters, at partitions, and at partition tables. I was not able to grasp the concept of partition tables, hence I had to do my research online, where I found WonderShare's article on the subject where they were able to provide me with an in-depth analysis of how they worked and why they were needed. We ended this part of the learning resource by looking at data layers, which explained the difference between allocated and unallocated data before we finally looked at HDD data transfer speed to end the slides.

Reflection on Seminar Discussions (week 8)

For this week, we were tasked with creating an infographic for the registers in an 8086 microprocessor, which were to be created at the request of the CM1205 module. We first looked at other infographics available online to see what we were aiming for. Through our research, we were able to notice a lot of bright colors across the board, with small icons representing the separate parts, which may be difficult in our case. Visual aids such as this, as mentioned previously, tend to help my learning, so I was intrigued by the idea of submitting my ideas for this design.

When we shared our ideas for the infographics in our seminar groups, it appeared we all had somewhat similar designs – 4 major sections, with bright colors and descriptions of the parts. We varied in how these were broken down – whether that be boxes or a circle. However, the general idea was the same. We all had this idea due to the similarity between the infographic and the actual build of the 8086 registers, and the need for simplicity of the design to create ease of learning for those who needed the aid. I used notes I had previously made on this subject to get the relevant information, whilst not having too much additional data.

WEEK 9: MEMORY, SECONDARY MEMORY, I/O, BUS CONTINUED

Reflection on Learning Materials

This was week 9 of learning for CM1205, in which we continued from the previous week's work on memory. As discussed before, with the storage hardware and some organization systems, I am fairly familiar with them due to my previous knowledge; however, we began this week by looking at the RAID system for HDD drives, a new subject for me. I understood to a certain level how the RAID system worked, however, it was "Dave's Garage" on YouTube who made a video titled "How RAID works" which gave me the full understanding I required. With this, I could continue and understand the 6 levels presented to us, explained with the use of diagrams as seen below:



RAID 0

These were examples of how different RAID organizations can be implemented to organize the disks. Each strip has several contiguous sectors. The difference between the two of these, however, is that RAID 1 provides a backup. These RAID organizations are vital to organizing data and minimizing the space required on the disk drives. Following HDD, we took a brief look at Solid State Drives (SSD). I learned about how the data was stored in pages and blocks and how these were used to read and write from, including "garbage collection" and the use of "TRIM".

Next, I learned about the last type of memory – optical. This was split into CD-ROM, DVD, and then Blu-ray. I used the resources available to compare the differences in how they store data and how we are (or not in the case of CD ROM) able to rewrite onto the disks, along with reading and the initial writing. As previously mentioned, I am somewhat familiar with how disks work so the information mentioning how the disks operate was not new to me. However, specifications such as the file systems in CDs were something I was able to discover. I am most familiar with the Joliet version of filing on CDs, as my primary devices are Windows-operated.

The final bit of learning I partook in this week required me to understand the I/O devices and data transfer. This was a fairly basic concept that I had once again seen before, hence I did not need too much explanation. Despite this, I was still able to learn from the data transfers between different I/O devices, how they use buses and the CPU, and require controllers to communicate. We specifically looked at the different busses in use by these I/O devices, given as follows:

- The ISA Bus
- The PCI (Peripheral Component Interconnect) Bus
- PCI-X (Extended)
- PCI Express (PCIe)
- SOUTHBRIDGE controller

Overall, I was still able to learn about the intricate details within the storage devices of the computer despite my previous knowledge, especially the communication between said devices.

Reflection on Seminar Discussion (Week 9)

For this seminar, we were required to discuss ideas under the scenario that we were attempting to gain insights from data organizations in computers to improve the organization of the company. We first looked at how in data organization, there is modularity in that each part is broken down into smaller parts such that computers can do parts individually before passing it on to the next piece that requires it. This is possible to implement in the organization as we can break down tasks in a way that allows optimal work to be done so each person does something separate before combining to solve the greater task, which is possible with humans.

However, an issue we discussed with these comparisons is that there are differences between humans and data, such as complexity like emotions and adaptability, such that some people may not be able to work as well with certain languages. Despite this, some key concepts we believe would still work. For example, a hierarchy system – a common system within computers - to separate managers of projects and teams would be good to implement within the company to organize the roles within to optimize the output of work. Another example of how we can implement data systems into teams of people is can encouraging performance reviews. Much like computers run diagnostics, we can review which parts of the company are meeting their targets, and what is working and what is not.

In conclusion, with an understanding of how computer systems work, we are quite easily able to implement changes in a company to improve efficiency—for example, reviews, modularity, and hierarchy.

WEEK 10: OPERATING SYSTEMS

Reflection on Learning Materials

This is week 10 of the CM1205 learning. This week, we began to look at operating systems – the resource management control program that runs every computer. The slides presented to us this week showed us how Operating Systems work at a higher level than I have learned before, despite my previous knowledge. We looked at the different types of computer systems: batch systems, seen in early computers; multi-programmed systems, having the CPU running tasks simultaneously; timesharing systems, multitasking systems that interact with multiple users; embedded systems, systems designed for one specific goal such as microwaves or traffic lights and finally distributed systems, essentially networks of systems. For distributed systems, we were encouraged to learn further through the use of this website to understand how Microsoft used this, as these systems were mainly developed for web browsing:

https://www.economist.com/business/1997/10/23/a-bundle-of-trouble

From this, we also had a quick look at networking, a subject that I am extremely familiar with. This was the end of the first segment on operating systems.

The next part consisted of scheduling. I explored a mix of cycles and processes used when coordinating between the CPU and I/O devices using techniques such as scheduling preemptively and non-preemptively. We were presented with explanations on many scheduling methods. These included:

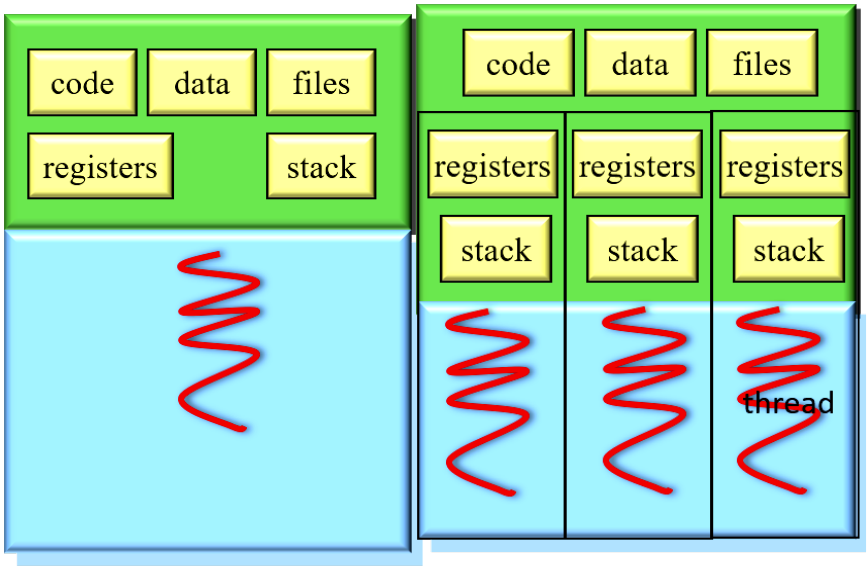
- First-Come, First-Served
- Shortest Job First
- Round Robin

All of these I am very familiar with due to my previous experience with computer systems; hence this week of learning required no extra research and was relatively easy to comprehend.

Reflection on Seminar Discussion (Week 10)

This week's seminar was to discuss how we would begin to manufacture NAND flash memory in the UK. Within my group, we discussed many factors and documented a brief analysis of the requirements from this project. Firstly, we discussed what NAND flash memory was – this is non-volatile data chips, for example USB sticks. Then, the first thing we looked at is resources needed to do create a process of manufacturing NAND chips on a large scale. Our example required us to look at the task from the perspective of the UK government, and in doing so we must take into account we have a decently large amount of money (in theory) and the availability of those whom are looking for jobs, which will be a high number of people. Next, we researched the actual components of the chips: water, irons, sand, a metal-oxide semiconductor material. These are generally resources available to us in the UK.

Then, we took into account the market. We made a list of producers currently in the market, with the three major competitors being: Samsung, Intel, Winbond. These are the companies whom currently control the market selling flash memory, meaning we would have to consider how we would convince new people looking for flash memory. The main two factors we would focus on would be price and quality as being able to beat all our competition in both of these means we wouldn't need to convince people that our product is good as ours would be the best factually. From this, we were able to build a general plan in order to show how we would develop a solution to the problem presented.



Single-threaded process Multi-threaded process

WEEK 11: PROCESSES AND THREADS

Reflection on Learning Materials

This was the final week of learning in this project for CM1205. We began by looking at what processes are with the slideshow presented

to us. A process is a program containing program code data, addresses, CPU registers, Program Counter and state. Alongside this was the discussion of Process Control Blocks and Process Queues which was able to expand this smaller simpler segment into a new concept for myself: threads. These are used for when single applications are required to perform several tasks. These are a sequence of instructions given to the CPU to improve performance, of which share code and data and open resources while having individual Program Counter values, stack pointers and stacks. This is something that I was not able to grasp the concept of once again, hence I was required to do my own research where I went to "ioflood.com" which gave me a small description worded differently to understand this. We were shown the differences between single and multi-threading in diagrams shown below:

There are generally more benefits to multithreaded programs, such as interactivity, resource sharing, speed and concurrency. This was the end of the first half of the lecture slide, which generally was not as difficult.

Process synchronizing was the second half of the slides in which we were discovered why we needed this in order to share processes. However, we have an issue in that we have a critical section where multiple threads can access the same data causing an overlap. I discovered we have beaten this issue by calling for permission to enter this section using sophomores, which run this check.

The final segment for this part of CM1205 was memory management, an area I am quite familiar with in my experience with computers and computer sciences. Despite this, we discussed how they were allocated in further detail than I had previously had knowledge about but with the same general ideas staying relatively similar.

Reflection on Seminar Discussion

Our task for this final week was to study the "tragedy of the commons" and compare it to CPU scheduling. We began by doing research into this "tragedy" in our group. We discovered this is where people act independently and spoil shared resources meaning that neither they nor the other party are able to access this. Then, we looked at CPU scheduling, which we have looked at in week 11 for our learning resource. This is the process of allocating CPU resources in order to maximize system work rate.

Next, we compared these two concepts. We compared how resources are shared and how they can be regulated. The general idea is that not everyone has access to all of the resources constantly that are available, and checks such as process synchronizing are used in order to make sure no two processes use the resource simultaneously. In conclusion, despite the two operating differently, they can be managed in similar ways.